Data Structures and Algorithms -3 (25CS2103E)

Large Prime Number Validation Using Miller–Rabin Algorithm

Team Members

Ratna Priya (2520030156)

Vaibhava Sree (2520030398)

Chandra Hasini (2520030086)

Team Number: 01 | Section: 05

Under the Esteemed Guidance of

D. Swathi

Professor, CSE Department

Contents

- Abstract
- Introduction
- Literature survey
- Problem Statement and Objectives
- Hardware and Software requirements
- Results
- Conclusion
- Future Scope

Abstract

- This project is a web-based service to check whether a very large number is prime.
- It uses the Miller–Rabin primality test, a randomized (probabilistic) algorithm.
- Trial division is too slow for very large numbers.
- The number is tested in many random rounds.
- If a witness is found, the number is Composite. Otherwise it is Probably Prime.
- The application shows the result, execution time and confidence level.

Introduction

- A prime number has only two factors: 1 and itself.
- Large prime numbers are used in RSA and other security systems.
- Trial division checks up to √n, so it is very slow for big numbers.
- Miller–Rabin is fast and gives the answer with very high confidence.
- Composite result is always correct. Probably Prime result has a very small error chance.
- Syllabus link: DSA-3 Module-1, large-prime primality testing (randomised algorithms).

Literature Survey

Method	Type	Drawback
Trial Division	Deterministic	Very slow for large numbers
Fermat Test	Probabilistic	Fails for Carmichael numbers like 561
Solovay–Strassen	Probabilistic	Error is higher (up to 1/2 per round)
Miller–Rabin	Probabilistic	Gives only Probably Prime
AKS Test	Deterministic	Too slow in practice

- We chose Miller–Rabin because it is fast, simple to implement and its error becomes very small as rounds increase.

Problem Statement and Objectives

Problem Statement

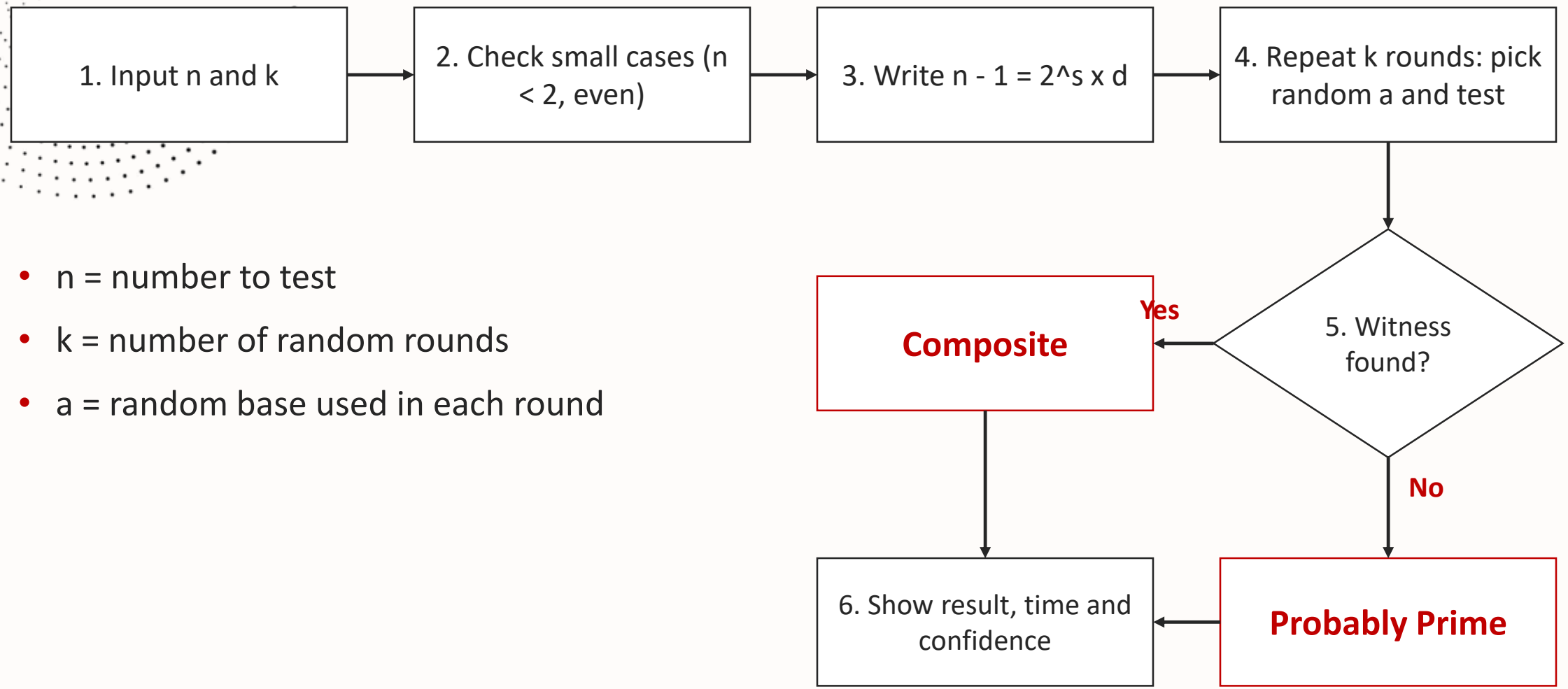
Checking whether a very large number is prime is slow with trial division.

We need a fast method that shows the result with a confidence level.

Objectives

- Build a web service to validate large numbers
- Implement the Miller–Rabin test
- Use random bases in multiple rounds
- Show result, time and confidence level
- Show that it is faster than trial division

Flow Chart



- n = number to test
- k = number of random rounds
- a = random base used in each round

Results

Input number	Digits	Output
97	2	Probably Prime
561	3	Composite
1105	4	Composite
2^31 - 1	10	Probably Prime
2^61 - 1	19	Probably Prime
2^127 - 1	39	Probably Prime
2^127 + 1	39	Composite
2^521 - 1	157	Probably Prime

Observations

- Tested with 20 rounds. All results are correct.
- 561 and 1105 are Carmichael numbers, and they are correctly found composite.
- Error chance after k rounds is at most 1/4^k.
- For 20 rounds it is less than 1 in 10^12.
- Trial division on 2^127 - 1 needs about 10^19 steps. Miller–Rabin needs only a few thousand multiplications.

Hardware and Software requirements

Category	Component	Requirement
Hardware	Device	Laptop / Desktop
	RAM	Minimum 4 GB
	Input	Keyboard and Mouse
	Network	Internet connection
Software	Operating System	Windows / macOS / Linux
	Language	Java
	Frontend	HTML, CSS, JavaScript
	Code Editor	VS Code
	Browser	Chrome / Edge

Conclusion

- The project validates very large numbers using the Miller–Rabin algorithm.
- It is much faster than trial division.
- A witness proves a number is composite.
- More rounds give higher confidence that a number is prime.
- The web application shows result, execution time and confidence level.
- It shows how a randomised algorithm can solve a real problem.

Future Scope

- Deterministic test for 64-bit numbers using fixed bases
- Combine with Baillie–PSW / Lucas test
- Generate large prime numbers for RSA keys
- Run the rounds in parallel using multiple threads
- Upload a file and test many numbers at once
- Deploy on the cloud with user login

GEO-TAG Photos





(DEEMED TO BE UNIVERSITY)

Thank you